Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and verify sequential storage elements and register-based data storage systems using programmable logic devices.	
Experiment No: 7	Date:	Enrolment No: 92301733013

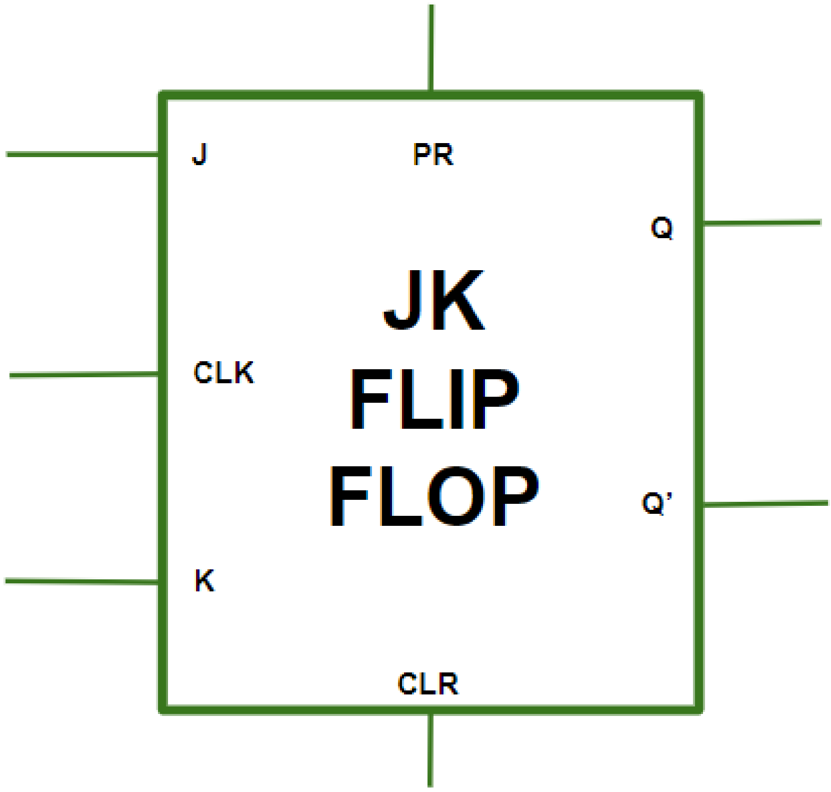
Aim: **To design and verify sequential storage elements and register-based data storage systems using programmable logic devices.**

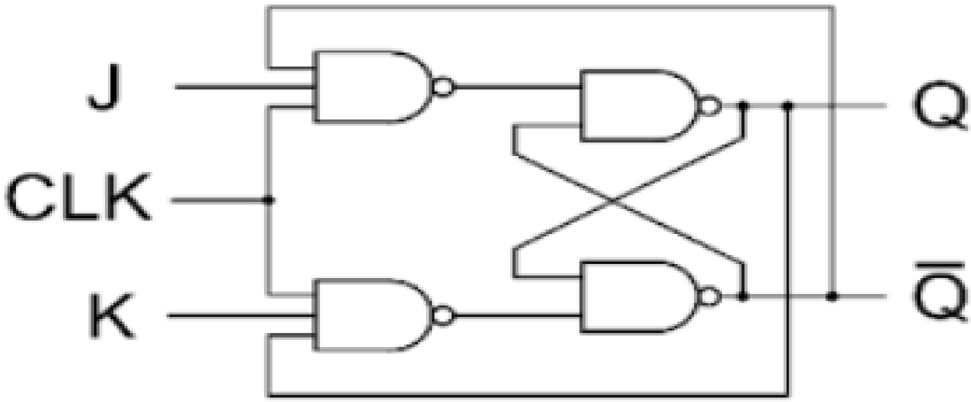
Software required: Xilinx Vivado

Hardware required: Nexys4 DDR FPGA Development Board

Theory:

JK Flipflop



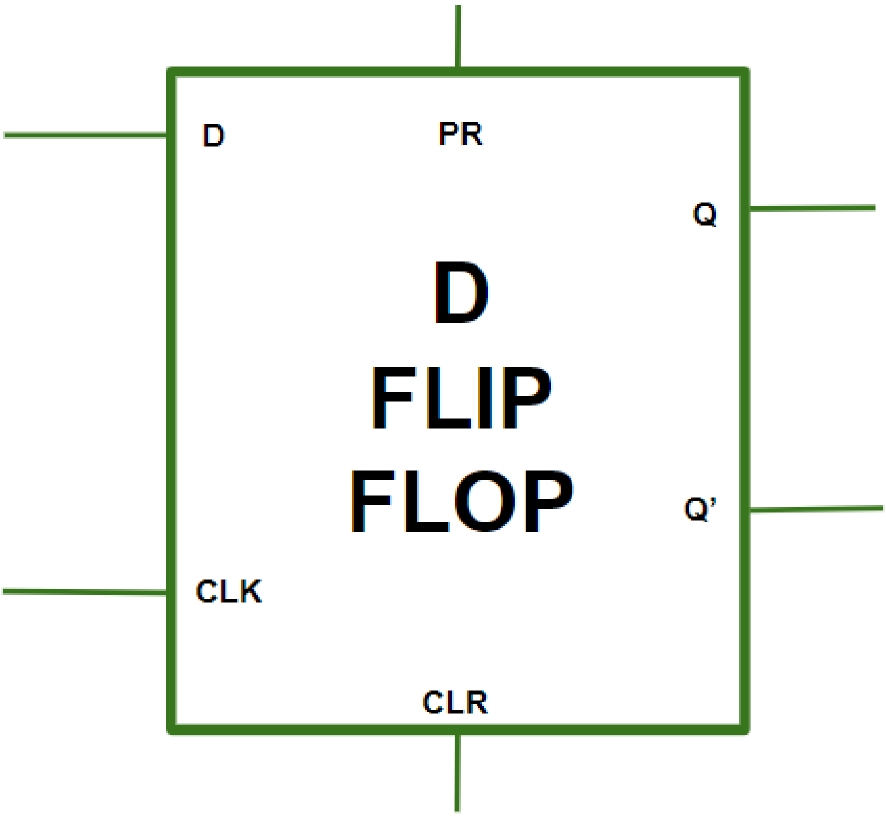


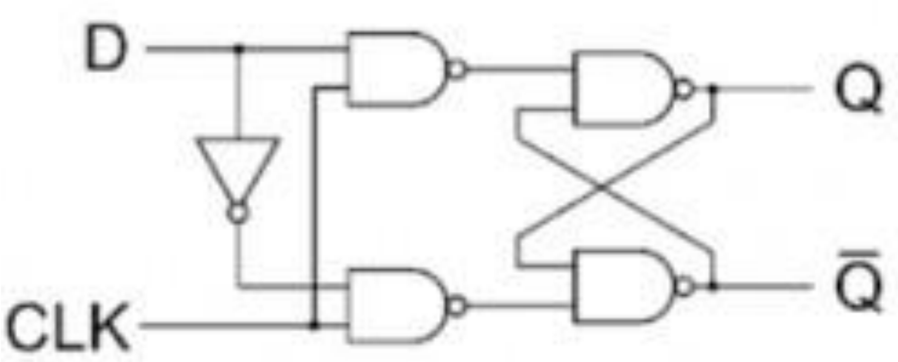
TRUTH TABLE

J	K	Q_N	Q_{N+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

- Case 1 (PR=CLR=0):This condition is in its invalid state.
- Case 2 (PR=0 and CLR=1):The PR is activated which means the output in the Q is set to 1. Therefore, the flip flop is in the set state.
- Case 3 (PR=1 and CLR=0):The CLR is activated which means the output in the Q' is set to 1. Therefore, the flip flop is in the reset state.
- Case 4 (PR=CLR=1):In this condition the flip flop works in its normal way whereas the PR and CLR gets deactivated.

D Flipflop

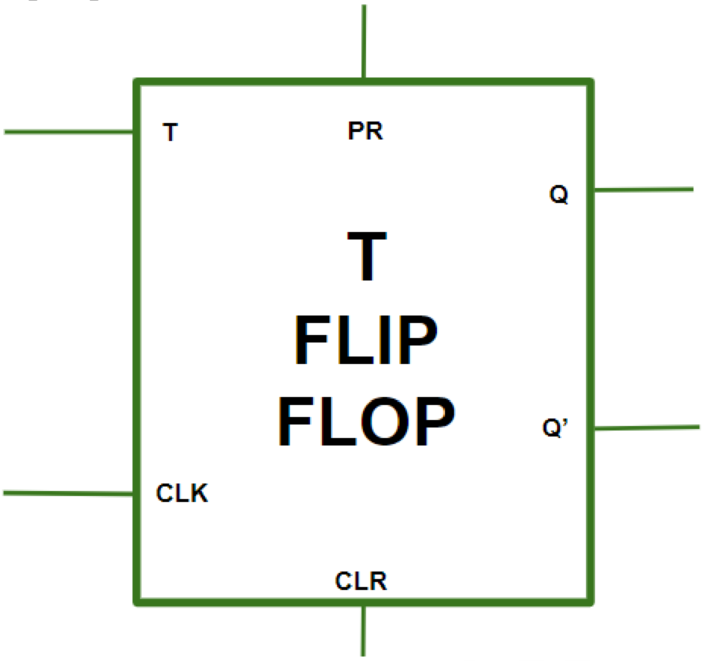


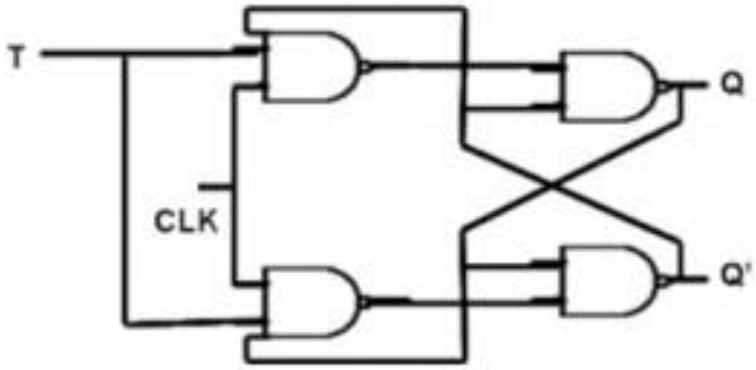


Q	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

- **Case 1 (PR=CLR=0):**This conditions is represents as invalid state where both PR(present) and CLR(clear) inputs are inactive.
- **Case 2 (PR=0 and CLR=1)**:This state is set state in which PR is inactive (0) and CLR is active(1) and the output Q is set to 1.
- **Case 3 (PR=1 and CLR=0):**This state is reset state in which PR is active (1) and CLR is inactive (0) and the complementary
- **Case 4 (PR=CLR=1):**In This state the flip flop behaves as normal, both PR and CLR inputs are active(1). output Q' is set to 1.

T FlipFlop





T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

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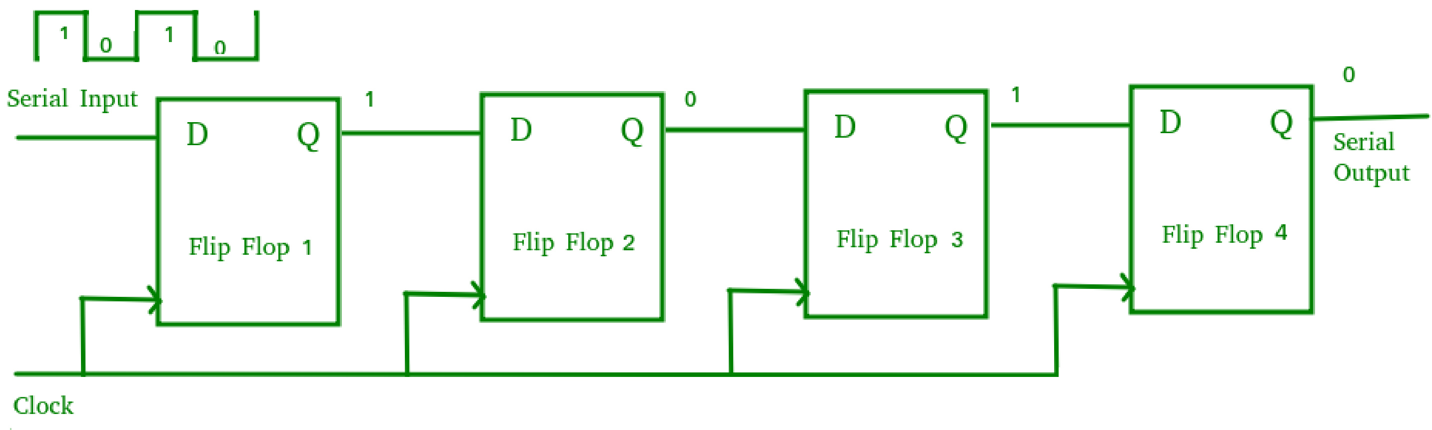
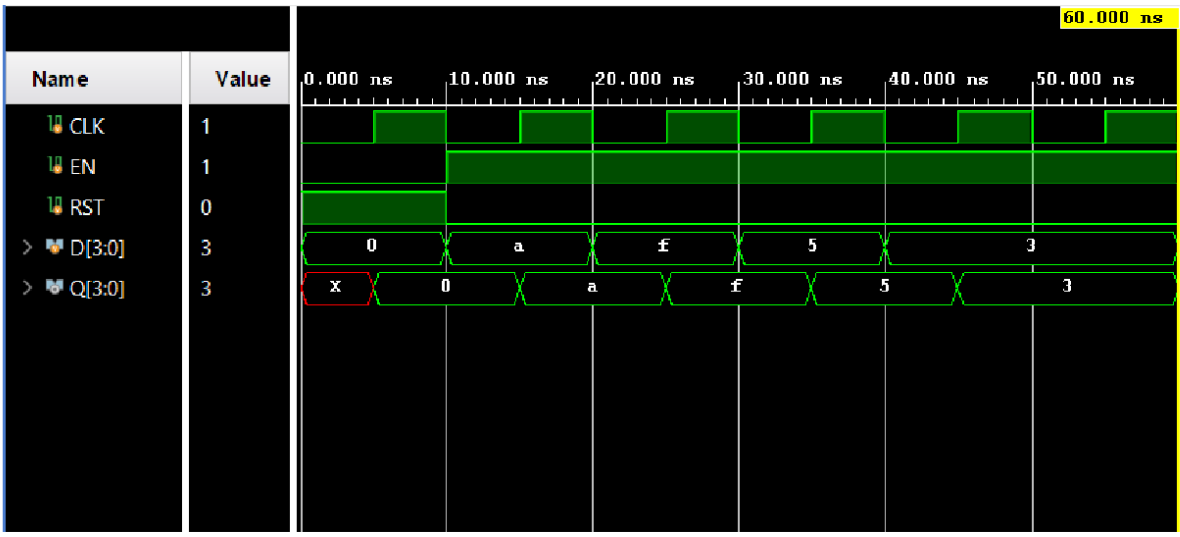
Register



Verilog Code:

```
module register4bit(
    input CLK,
    input [3:0] D,
    output reg [3:0] Q
 );
    always @(posedge CLK)
      begin
         Q <= D;
      end
endmodule
```

Simulation Results:



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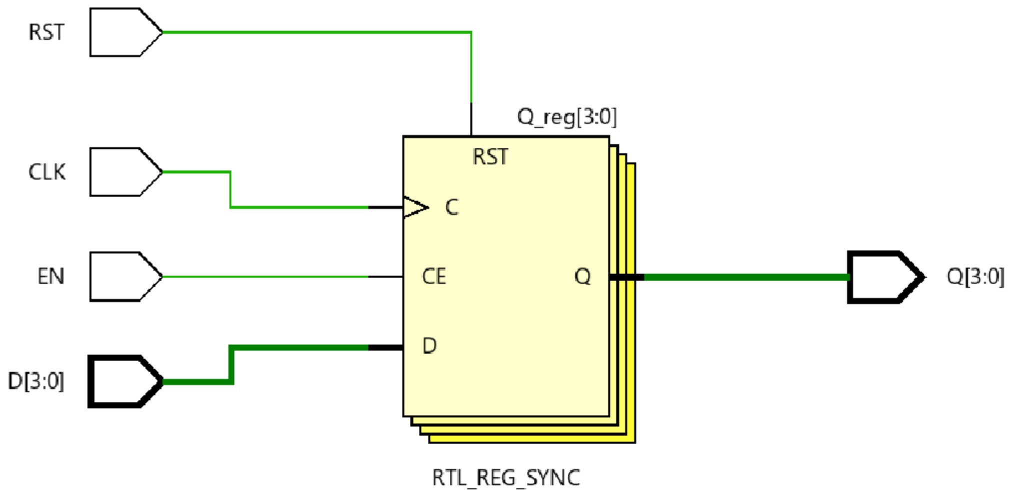

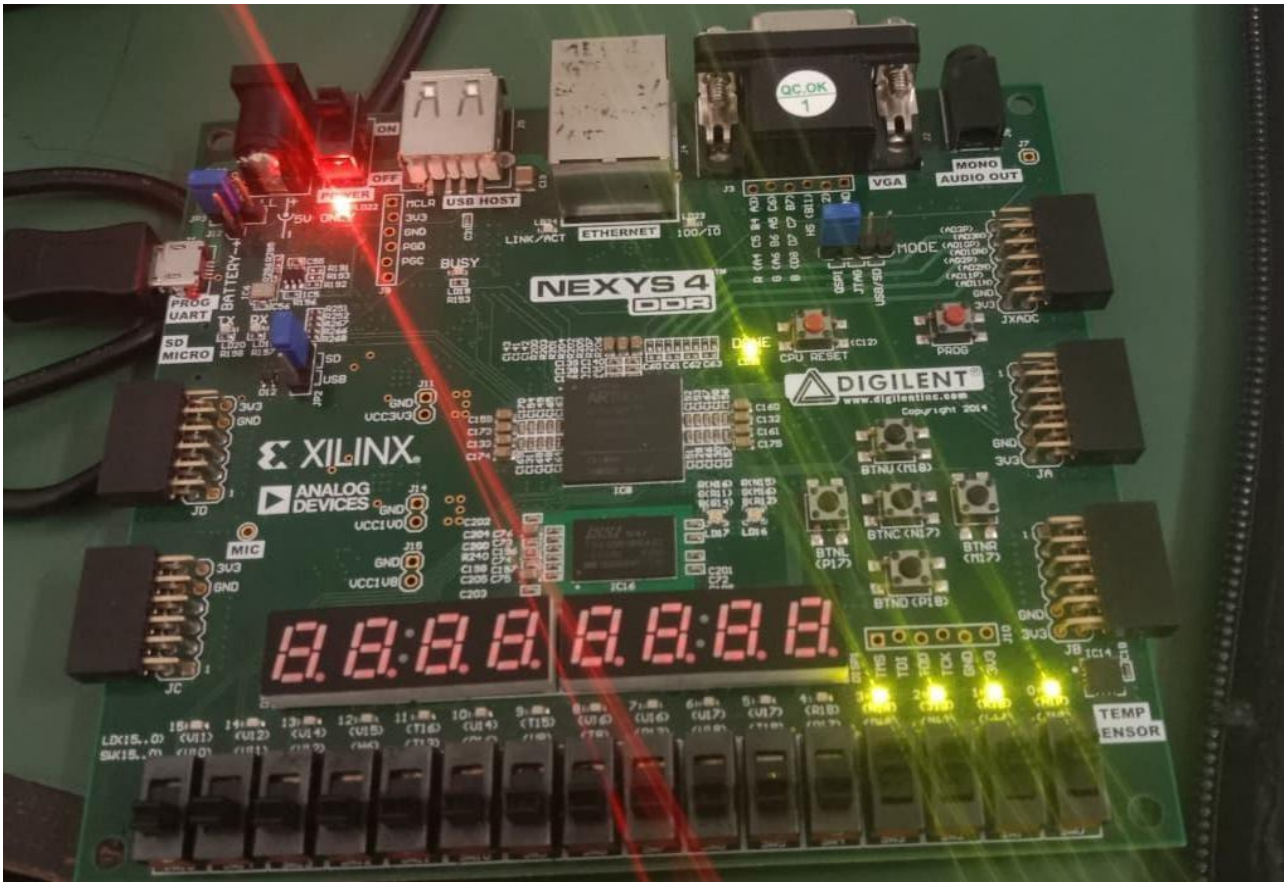

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Conclusion:

The flip-flop and register circuits were successfully designed, simulated, and implemented on the FPGA development board using Verilog HDL. The simulation and hardware results confirmed that the sequential storage elements and data storage functions operated as expected. This experiment provided a better understanding of clock-driven sequential circuits, timing behavior, and the practical implementation of digital storage systems on an FPGA.

Post Lab Exercise:

1. Which signal controls the operation of a flip-flop?The clock (CLK) signal controls the operation of a flip-flop. The flip-flop changes its output only on the rising or falling edge of the clock, depending on its design .
2. Why are registers used in digital systems?
   Registers are used to store and transfer binary data temporarily. They are essential for data storage, buffering, arithmetic operations, and communication between different parts of a digital system .

3. What is the purpose of the reset input in a flip-flop?
   The reset input initializes the flip-flop by setting its output to a known state (usually 0). This ensures the circuit starts operating correctly and can recover from unwanted states.

4. Why is a clock edge important in sequential circuits?

   The clock edge synchronizes all sequential operations. It ensures that data is captured and outputs change only at specific instants, preventing unpredictable behavior.

5. What How does setup and hold time affect the reliability of flip-flop operation?

   The input data must be stable before the clock edge (setup time) and remain stable after the clock edge. Violating these timing requirements can cause metastability and incorrect outputs, reducing circuit reliability.

6. Why is synchronous reset generally preferred over asynchronous reset in FPGA designs?

   A synchronous reset changes the state of a flip-flop only at the active clock edge. This makes timing easier to manage and helps minimize unwanted glitches. As a result, synchronous resets provide more reliable operation and are well suited for FPGA-based designs.

7. What is the effect of using non-blocking assignments in Verilog sequential logic?

   Non-blocking assignments (<=) update all registers simultaneously at the end of the clock cycle, accurately modeling hardware behavior and preventing race conditions in sequential circuits.